Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy whose feasibility screenings led a client to approve the top-ranked site ($1.5M projected yearly revenue).
- Built a revenue driver tree (traffic × capture rate × average ticket) that turned owners' profit goals into required car-count targets, with low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR, CVaR).
- Estimated $533K in annual incremental county revenue per added car wash with Double ML (econml, XGBoost nuisance models) on a 234-county panel (2012-2023), cross-checked with Callaway-Sant'Anna DiD ($472K) and a synthetic-data recovery test.
- Engineered Python ETL pipelines with automated data-quality checks and 250+ unit tests for Census, traffic, and competitor data, cutting market analysis turnaround by 91% (weeks to days).
- Designed a site-scoring model combining demographic, traffic, competitive-density, and consumer-behavior features, explained to owners with uncertainty ranges in written reports and maps.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, finance, engineering, and behavioral science leadership, setting OKRs around the core KPIs: gross lead rate (GLR), enrollment rate, and cost per member.
- Identified a client's "Opportunist" segment as the main driver of GLR, pivoting targeting toward it and reallocating 12% of campaign budget.
- Applied predictive, segmentation, and uplift models to millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible consumer records, adopted in 90% of campaigns and cutting spend by $10K (about 3%) per campaign at baseline enrollment.
- Built an in-house experimentation platform that automated A/B test design and readouts for direct-mail campaigns (power analysis, CUPED, significance tests), with CUPED cutting metric variance 20% so tests needed smaller samples.
- Re-architected the production churn prediction pipeline in Python, SQL, and Spark on Databricks, cutting inference runtime 48% and memory 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation frameworks, and drift monitoring for production search and regression models, validated offline and with A/B tests, and trained the 16-person product and engineering team on ML.
- Built SQL training tables that broke bulk orders into item-level shipping costs, then deployed a regression-based shipping-price API in C# on Azure, cutting median price error vs. carrier cost from $13 to $4 on standard orders while maintaining contribution margin.
- Partnered with product and engineering to cut product search dissatisfaction scores 12% and raise click-through rate by tuning Elasticsearch relevance (custom analyzers, field boosting), validated with NDCG@10, human review, and A/B tests.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built and validated XGBoost classification models on millions of healthcare consumer records with hundreds of external attributes, raising offline AUC-ROC from 0.62 to 0.67, explaining drivers with SHAP, and confirming a 9% GLR lift in an A/B test.
- Standardized the core campaign metrics (gross lead rate, enrollment rate, cost per member), resolving conflicting GLR definitions with data engineering for a client missing key lead markers, and validated GLR out of sample as a 60-day leading indicator of enrollment rate ($R^2 = 0.49$).
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, including a company-wide self-service dashboard.
- Doubled the precision of high-opportunity zone selection (15% to 30%, measured against past successful campaigns) with GeoPandas spatial analysis.
- Owned the monthly model monitoring report covering drift, performance, and forecast-vs-actual accuracy for up to 5 production models.

TECHNICAL PROJECTS

CaLaun — *AI-Powered Calculus Tutor (LLM Application)* github.com/calaun-learning/calaun

- Built and deployed a generative AI tutor that grounds a self-hosted Llama 3.2 LLM in SymPy-verified solution steps, scoping each prompt to one step so the model explains but never computes.

Medicare Claims Intelligence github.com/beauvilerobed/.../projects/medicare-claims-intelligence

- Built a dbt warehouse with 26 data tests over 790K synthetic CMS Medicare claims and a fraud (FWA) detector checked against planted billing schemes.
- Forecast weekly flu-like-illness visits (CDC ILINet) with 41% lower error than a seasonal-naive baseline (MASE 0.62 vs. 1.06), and built a SHAP-explained XGBoost model that finds 46% more next-year high-cost members than ranking by prior-year cost.

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/.../projects/ltv-churn-survival

- Simulated 1M customer lifetimes to show where LTV = ARPU / churn fails, estimating customer lifetime value from empirical survival curves.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, statsmodels, econml, GeoPandas), SQL, C#, Bash, PowerShell

Machine Learning & AI: Regression, Classification, Clustering, XGBoost, Anomaly Detection, Time Series Forecasting (ARIMA, ETS, Prophet), Survival Analysis, Uplift Modeling, SHAP, NLP, LLMs (Llama), Prompt Engineering, Feature Engineering, Model Monitoring & Drift

Statistics & Experimentation: KPI Frameworks, Driver Trees, Metric Definitions, Leading Indicators, Hypothesis Testing, A/B Testing, CUPED, Power Analysis, Causal Inference, Double ML, Causal Forests, Difference-in-Differences, CATE, Monte Carlo Simulation, Optimization

Data Engineering & Cloud: Spark, Databricks (Workflows), Snowflake, dbt, MLflow, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT Pipelines, Data Quality Checks, Unit Testing, Git, AI-Assisted Development (Claude Code)

Visualization & Communication: Power BI, Tableau, Excel, Matplotlib, Plotly, Dashboards, Stakeholder Presentations, Mentoring